

Scalability & Future Plan

Date	5 july 2026
Team ID	
Project Name	HDI Prediction System using Machine Learning and Flask
Maximum Marks	1 Mark

Scalability & Future Plan:

The HDI Insight AI system was developed as a prototype for predicting the Human Development Index using machine learning. The current application works efficiently for individual predictions through a Flask web interface. In the future, the system can be enhanced to support larger datasets, multiple users, improved security, cloud deployment, and more advanced machine learning models for higher prediction accuracy.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only single-user access	Limited scalability	High
2	Uses a Linear Regression model	Prediction accuracy can be improved	Medium
3	Local deployment only	Cannot be accessed remotely	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports one user at a time	Deploy on cloud with multi-user support
2	Data Storage	CSV dataset	Use MySQL or PostgreSQL database
3	Performance	Basic Linear Regression	Use Random Forest or XGBoost for better accuracy

4	Security	Basic Flask application	Add user authentication and HTTPS security
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Deploy application on Render/AWS	1 Month	Online accessibility
Phase 2	Database integration	2 Months	Online accessibility
Phase 3	Improve ML model accuracy	3 Months	More reliable predictions

Phase 4	User login, dashboard, and analytics	4 Months	Enhanced user experience
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